

1. Administrative & Study Identification

**Full Study Title:** A Study of Zilovetamab Vedotin (MK-2140) as Monotherapy and in Combination in Participants With Aggressive and Indolent B-cell Malignancies (MK-2140-006) **Short Title/ Acronym:** waveLINE-006 **Protocol Number:** MK-2140-006 **NCT Number:** NCT05458297 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Zilovetamab vedotin (MK-2140) alone and in combination with nemtabrutinib **Phase of Development:** Phase II **Medical Monitor:** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To assess the safety and tolerability of zilovetamab vedotin as monotherapy and in combination with nemtabrutinib, and to evaluate the objective response rate (ORR). **Secondary Objectives:** To evaluate the duration of response (DOR), progression-free survival (PFS), overall survival (OS), and additional safety and tolerability measures of the regimens.

**2.2 Background & Rationale** To address the unmet medical need for improved treatments in heavily pretreated relapsed/refractory (R/R) B-cell malignancies. ROR1 is a transmembrane protein expressed in hematologic malignancies. Zilovetamab vedotin (ZV) is an antibody-drug conjugate comprising a humanized IgG1 monoclonal anti-ROR1, a proteolytically cleavable linker, and the antimicrotubule agent monomethyl auristatin E. The combination of ZV and nemtabrutinib (a reversible BTK inhibitor) has the potential for improved, synergistic responses in selected B-cell malignancies.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Single-arm (for monotherapy cohorts) and partially randomized (for schedule optimization) **Blinding:** Open-label **Randomization:** 1:1 (only in Cohort D for ZV schedule optimization)

**3.2 Planned Enrollment & Duration Target \$N\$:** Approximately 275 participants across 6 cohorts (A-F) **Number of Sites:** Global multi-center (including US, Italy, Czech Republic, Israel, Chile, etc.) **Estimated Study Start:** 2022 **Estimated Primary Completion:** Ongoing

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age  $\geq 18$  years with biopsy-proven and/or histologically confirmed mantle cell lymphoma (MCL), Richter's transformation (RT), chronic lymphocytic leukemia (CLL), or follicular lymphoma (FL). 2. Relapsed or refractory (R/R) disease after at least 2 prior systemic therapies (for MCL, must include a BTK inhibitor and be post-CAR-T or ineligible). 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 and adequate organ function.

**4.2 Exclusion Criteria** 1. Known active central nervous system (CNS) lymphoma involvement. 2. Prior radiotherapy within 28 days of the start of study intervention or ongoing corticosteroid therapy exceeding 30 mg daily of prednisone equivalent. 3. Active infection requiring systemic therapy, or active HBV or hepatitis C virus (HCV) infection.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Zilovertamab vedotin 2.5 mg/kg IV Q3W or 2.0 mg/kg IV Q2/3W (determined via schedule optimization). For combination cohorts, ZV is administered with oral nemtabrutinib. **Route of Administration:** Intravenous (IV) for ZV, Oral for nemtabrutinib. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or other predefined discontinuation criteria are met.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard supportive care therapies. **Prohibited:** Concurrent investigational anti-cancer therapy, systemic corticosteroids  $>30$ mg/day prednisone equivalent, and live or live-attenuated vaccines within 30 days before the first dose.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                        |
|--------------|---------------|----------------------------------------------------------------------------------|
| Screening    | Day -28 to 0  | Informed Consent, Medical History, Baseline tumor scans                          |
| Baseline     | Day 0         | Enrollment, First Dose                                                           |
| Interim      | Q2W or Q3W    | IV infusions, Safety Labs, Tumor scans Q12W up to week 108, then Q24W thereafter |
| End of Study | Variable      | End of Treatment (EOT) Visit, Safety Follow-Up, Survival Follow-up               |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Safety and tolerability profile (AEs, DLTs); and Objective Response Rate (ORR), defined as the percentage of participants who achieve a complete response (CR) or partial response (PR). **Measurement Method:** AEs graded per NCI

CTCAE version 5.0. Response assessed by Blinded Independent Central Review (BICR) and investigator per Lugano Response Criteria.

**7.2 Secondary & Exploratory Endpoints** 1. Duration of Response (DOR) 2. Progression-Free Survival (PFS) 3. Overall Survival (OS)

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** AEs will be continuously monitored and graded per NCI CTCAE version 5.0. **Serious Adverse Events (SAEs):** Continuously tracked; dose-limiting toxicities (DLTs) assessed specifically during safety run-in cohorts. **Data Monitoring Committee (DMC):** Internal sponsor review committee to evaluate safety data and recommend Recommended Phase 2 Dose (RP2D).

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) population for efficacy outcomes; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Target  $N = 275$  is powered to evaluate the safety run-in combinations, schedule optimization (randomized 1:1), and efficacy expansion across multiple indication-specific cohorts. **Handling of Missing Data:** Participants with missing response assessments are considered non-responders per standard SAP missing data provisions.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote sponsor monitoring throughout the treatment phase. **Audit Trail:** Assured via eCRF locking and independent centralized review of tumor scans (BICR).

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## 11. Study Identifiers & Governance

**Primary ID:** MK-2140-006 **Registry Number:** NCT05458297 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study of Zilovetamab Vedotin (MK-2140) as Monotherapy and in Combination in Participants With Aggressive and Indolent B-cell Malignancies (MK-2140-006) **Official Regulatory Title:** A phase 2 study of the safety and efficacy of zilovetamab vedotin as monotherapy or in combination in patients (pts) with aggressive and indolent B-cell malignancies

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## 12. Clinical Framework (Oncology Specific)

**Primary Condition:** Aggressive and Indolent B-cell Malignancies (MCL, RTL, FL, CLL) **Diagnosis Threshold:** Histologically confirmed biopsy of relapsed or refractory disease **Medical Methodology:** ROR1-targeted Antibody-Drug Conjugate (ADC) combined with BTK inhibition **Optimization Target:** Objective response rate (ORR) and safety tolerability

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## 13. Study Procedures & Methodology

**Management Pathway:** Monotherapy systemic IV infusion or combination with oral targeted therapy **Education Protocol (HCP):** Protocol-specific safety, DLT monitoring, and dosing guidelines **Education Protocol (Patient):** AE monitoring and reporting instructions, clinic visit adherence **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of imaging/tumor scans

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Follow-up until disease progression, unacceptable toxicity, or death **Onsite Visit Cadence:** Every 2 to 3 weeks corresponding with the IV dosing schedule **Remote Monitoring Frequency:** Continuous pharmacovigilance follow-up **Checklist Review Interval:** Tumor scans every 12 weeks (Q12W) up to week 108, then every 24 weeks (Q24W) thereafter

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|------|------|------|------|
| Role                     | Name   | Signature | Date          | :--- | :--- | :--- | :--- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |      | Lead |      |      |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |